

Neuro NBME 1

For each patient with weakness, select the most likely diagnosis.

- | | |
|--|--|
| A) Amyotrophic lateral sclerosis | F) Myasthenia gravis |
| B) Diabetic peripheral neuropathy | G) Myasthenic (Lambert-Eaton) syndrome |
| C) Duchenne muscular dystrophy | H) Myotonic dystrophy |
| D) Guillain-Barré syndrome | I) Polymyositis |
| E) Muscle phosphorylase deficiency (McArdle disease) | J) Subacute combined degeneration of the spinal cord |

1. A 56-year-old man comes to the physician because of a 6-month history of widespread muscle cramping. Examination shows a right footdrop, diffuse muscle atrophy, and fasciculations. Deep tendon reflexes are brisk. Sensory examination shows no abnormalities.

For each patient with weakness, select the most likely diagnosis.

- | | |
|--|--|
| ☐ A) Amyotrophic lateral sclerosis | ☐ F) Myasthenia gravis |
| ☐ B) Diabetic peripheral neuropathy | ☐ G) Myasthenic (Lambert-Eaton) syndrome |
| ☐ C) Duchenne muscular dystrophy | ☐ H) Myotonic dystrophy |
| ☐ D) Guillain-Barré syndrome | ☐ I) Polymyositis |
| ☐ E) Muscle phosphorylase deficiency (McArdle disease) | ☐ J) Subacute combined degeneration of the spinal cord |

2. A 64-year-old woman comes to the physician because of a 6-week history of aching of her shoulders and thighs and difficulty climbing stairs. Examination shows moderate weakness and tenderness of the proximal arm and leg muscles.

3. An 11-year-old boy with hemophilia A is brought to the physician because of headaches and decreasing school performance since falling 10 feet to the ground 2 weeks ago. He did not lose consciousness at the time and remembers striking his head. A CT scan of the head showed no abnormalities. Current examination shows bilateral papilledema and hyperreflexia on the left. Vital signs are within normal limits. Which of the following is the most likely diagnosis?

- ☐ A) Brain abscess
- ☐ B) Epidural hematoma
- ☐ C) Postconcussive syndrome
- ☐ D) Subarachnoid hemorrhage
- ☐ E) Subdural hematoma

4. A previously healthy 27-year-old man is brought to the emergency department because of a 3-day history of progressive weakness and shooting pains in the lower extremities and a 6-hour history of difficulty rising from a chair and climbing stairs. His temperature is 37°C (98.6°F). Muscle strength in the lower extremities is markedly decreased proximally and distally; there is weakness of the intrinsic hand muscles bilaterally. Deep tendon reflexes are absent throughout. Which of the following is the most likely diagnosis?
- ☐ A) Amyotrophic lateral sclerosis
 - ☐ B) Guillain-Barré syndrome
 - ☐ C) Lumbar spinal stenosis
 - ☐ D) Poliomyelitis
 - ☐ E) Polymyositis
5. A 77-year-old woman comes for a routine health maintenance examination. She states that occasionally she has difficulty remembering the names of her friends and that she does not sleep well and awakens early in the morning. She has also had increasing stiffness with walking over the past 10 years. She has hypertension treated with hydrochlorothiazide. Her blood pressure is 138/88 mm Hg. Physical and mental status examinations show no abnormalities. Which of the following is the most likely cause of this patient's symptoms?
- ☐ A) Dementia, Alzheimer type
 - ☐ B) Medication adverse effect
 - ☐ C) Multi-infarct (vascular) dementia
 - ☐ D) Parkinson disease
 - ☐ E) Normal aging
6. A 32-year-old woman is brought to the emergency department because of a 3-day history of dizziness and loss of balance resulting in multiple falls. Four years ago, she received the diagnosis of multiple sclerosis. She also has rheumatic valvular disease, atrial fibrillation, and mitral valve prolapse. Current medications include warfarin and interferon beta-1a. On arrival, her speech is dysarthric. She is 165 cm (5 ft 5 in) tall and weighs 67 kg (148 lb); BMI is 25 kg/m². Her pulse is 78/min and irregular, and blood pressure is 110/70 mm Hg. The pupils are equal, round, and reactive to light. The right pupil dilates when the flashlight is moved from the left to the right eye. Nystagmus is present on gaze to the right and left. Muscle strength is 5/5 in the upper extremities and 4/5 in the lower extremities. Muscle tone is increased in the lower extremities. Deep tendon reflexes are normal in the upper extremities and brisk in the lower extremities. Babinski sign is present bilaterally. Sensation to cold is decreased over the right side of the face, and sensation to cold and vibration is decreased below T4. Her gait is spastic and unsteady. Which of the following is the most appropriate next step in pharmacotherapy?
- ☐ A) Increase the dose of interferon beta-1a
 - ☐ B) Begin oral azathioprine
 - ☐ C) Begin subcutaneous glatiramer
 - ☐ D) Begin intravenous cyclophosphamide
 - ☐ E) Begin intravenous methylprednisolone

7. An unkempt 49-year-old man is brought to the emergency department in a confused and lethargic state. No history is obtainable. The sclerae are icteric. When his arms are held extended, irregular lapses of postural tone are observed. Test of the stool for occult blood is positive. Which of the following is the most likely diagnosis?

- ☐ A) Acute alcohol intoxication
- ☐ B) Epidural hematoma
- ☐ C) Hepatic encephalopathy
- ☐ D) Uremic encephalopathy
- ☐ E) Wernicke encephalopathy

8. A 37-year-old woman is brought to the emergency department by a coworker 45 minutes after she had a generalized tonic-clonic seizure at her office. Her coworker says that during the seizure she appeared to lose consciousness, her eyes and head deviated to the right, and she fingered her clothes peculiarly. On arrival, she is confused. Her temperature is 38.5°C (101.3°F), pulse is 120/min, and blood pressure is 150/95 mm Hg. Examination shows a left hemiparesis and neglect of the left side of the body. Deep tendon reflexes are 3+ in the left upper and lower extremities. Babinski sign is present on the left. On mental status examination, she is disoriented to person, place, and time. EEG shows repetitive complexes over the right temporal region. A CT scan of the head without contrast shows no abnormalities. A lumbar puncture is performed. Cerebrospinal fluid (CSF) analysis shows:

Pressure, opening	230 mm H ₂ O
Protein	80 mg/dL
RBC	652/mm ³
WBC	53/mm ³ (mostly lymphocytes)

Which of the following is most likely to confirm the diagnosis?

- ☐ A) Repeat CT scan of the head with contrast
- ☐ B) MRI of the brain
- ☐ C) CSF polymerase chain reaction test for herpes simplex virus
- ☐ D) Determination of CSF bacterial latex agglutinin antigens
- ☐ E) Determination of CSF cryptococcal antigen

9. A 47-year-old woman comes to the physician because of daily "crawling tightness" in her calves for 5 years. She notices the discomfort when she sits for long periods or lies in bed. The discomfort prevents her from falling asleep quickly. She reports that if she stands and walks, she obtains temporary relief. She is otherwise healthy and takes no medications. Peripheral pulses are normal. Neurologic examination shows no abnormalities. Which of the following is the most appropriate pharmacotherapy?

- ☐ A) Carbamazepine
- ☐ B) Clopidogrel
- ☐ C) Naproxen
- ☐ D) Pramipexole
- ☐ E) Quinine
- ☐ F) No pharmacotherapy is indicated

10. A 52-year-old woman comes to the physician because of constant severe burning pain over the right side of her ribs for 2 days. She says the pain keeps her awake at night and does not respond to acetaminophen or aspirin. During this period, she has had a mild rash over the painful area and intermittent low-grade fever. She has type 2 diabetes mellitus controlled with diet and hypertension controlled with a thiazide diuretic. Vital signs are within normal limits. Examination shows an 8-cm area of painful erythema and vesicles from T4 to the right costal margin. Results of a complete blood count and serum electrolyte concentrations are within the reference range. Which of the following is the most appropriate pharmacotherapy?

- ☐ A) Oral acyclovir
- ☐ B) Oral amantadine
- ☐ C) Oral ganciclovir
- ☐ D) Intravenous acyclovir
- ☐ E) Intravenous foscarnet
- ☐ F) Intravenous ganciclovir

11. A 35-year-old man has had nausea, vomiting, ringing of the ears, and a widespread rash for 10 days. He is an avid gardener. Examination of the skin shows seven 3-cm, brightly erythematous oval patches on the neck. There is increased pigmentation, especially around the axillae that have macules of depigmentation. Neurologic examination shows decreased sensation to pinprick over the feet. Ankle jerk reflexes are absent bilaterally. A photograph of the patient's fingernails is shown. Which of the following is the most likely diagnosis?



- ☐ A) Arsenic poisoning
- ☐ B) Pemphigus
- ☐ C) Porphyria variegata
- ☐ D) Psoriasis
- ☐ E) Secondary syphilis

12. A 32-year-old woman comes to the physician because of a 5-month history of progressive weakness and numbness of both legs. On examination, cranial nerve function is intact. Muscle strength is 4/5 in both lower extremities. There is numbness to pain and temperature along the lateral aspect in both feet up to the ankles. She has impaired two-point discrimination in her fingertips. Ankle and knee reflexes are absent; all other reflexes are intact. There is no atrophy. Analysis of cerebrospinal fluid shows a glucose concentration of 40 mg/dL and protein concentration of 120 mg/dL. Nerve conduction studies show slowed conduction velocity. Which of the following is the most likely structure involved in the underlying abnormality?
- ☐ A) Axons
 - ☐ B) Dystrophin
 - ☐ C) Myelin
 - ☐ D) Neurons in the anterior horn
 - ☐ E) Neurons in the caudate
 - ☐ F) Neurons in the motor cortex and corticospinal tract
 - ☐ G) Neurons in the substantia nigra
13. A previously healthy 32-year-old woman has the abrupt onset of a severe bifrontal headache, nausea and vomiting, and diplopia 24 hours after an uncomplicated vaginal delivery of a healthy 3175-g (7-lb) newborn. She also has had difficulty breast-feeding. She is drowsy but opens her eyes and moves all extremities to command. Her temperature is 37°C (98.6°F), pulse is 110/min, respirations are 15/min, and blood pressure is 115/60 mm Hg. Examination shows bitemporal hemianopia and weakness of the lateral rectus muscles bilaterally. The right pupil is 5 mm and reacts sluggishly to light directly and consensually; the left pupil is 3 mm and reactive. There is mild papilledema bilaterally and no meningismus. She is oriented to person and place but not to time. Which of the following is the most likely diagnosis?
- ☐ A) Brain stem infarction
 - ☐ B) Migraine
 - ☐ C) Pituitary apoplexy
 - ☐ D) Sagittal sinus thrombosis
 - ☐ E) Subarachnoid hemorrhage

14. A previously healthy 32-year-old man comes to the emergency department because of progressive left-sided low back pain during the past week. He works in construction and says that the pain began suddenly while he was working. The pain has not responded to acetaminophen. He appears uncomfortable and holds his back rigid when he walks. His temperature is 37°C (98.6°F), pulse is 76/min, respirations are 14/min, and blood pressure is 122/80 mm Hg. Physical examination shows paravertebral muscle spasm in the lumbar area and increased pain during straight-leg raising. Muscle strength is 5/5, and deep tendon reflexes are within normal limits. Sensory examination shows no abnormalities. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Complete blood count, serum chemistries, and erythrocyte sedimentation rate
- ☐ B) X-rays of the lumbosacral spine
- ☐ C) MRI of the lumbosacral spine
- ☐ D) Myelography
- ☐ E) No diagnostic studies are indicated

15. A 52-year-old woman is brought to the physician by her husband because of a 5-month history of personality changes. Her husband says that she has become irritable and believes that her neighbor is trying to sabotage her legal practice. She is also forgetful and has made several significant errors with the household finances. Her husband has also noticed unusual jerking movements in her arms and twitching of her face. Her father had dementia in his late 50s, but no other details of his illness or death are known. Examination shows facial grimacing, difficulty maintaining tongue protrusion, and difficulty with smooth pursuits. Cranial nerves are otherwise intact. There are intermittent rapid asymmetric jerking movements of the extremities. Muscle strength is full. Sensation is intact. There is no ataxia. Her Mini-Mental State Examination score is 20/30 with errors in recall and serial sevens. A CT scan of the head shows bilateral atrophy of the caudate nuclei. Which of the following is the most likely mechanism of these findings?

- ☐ A) Accumulation of cerebral cortical plaques and tangles
- ☐ B) Degeneration of the dopaminergic neurons of the substantia nigra
- ☐ C) Degeneration of striatal γ -aminobutyric acid neurons
- ☐ D) Diffuse cerebral infection with HIV
- ☐ E) Disseminated cerebral white matter demyelination
- ☐ F) Multiple cerebral infarcts

16. A 67-year-old man is brought to the emergency department 20 minutes after his wife found him unconscious in a pool of vomitus. He has hypertension treated with atenolol. He withdraws the left extremities vigorously to painful stimuli, but there is no movement of the right arm or leg. He moans incomprehensibly and opens his eyes briefly to noxious stimuli. His temperature is 37°C (98.6°F), pulse is 60/min, respirations are 10/min, and blood pressure is 180/95 mm Hg. The eyes deviate conjugately to the left. The left pupil is 6 mm and nonreactive, and the right pupil is 3 mm and reactive to light. Funduscopic examination shows no abnormalities. Deep tendon reflexes are 2+ in the left extremities and absent in the right extremities. Babinski sign is present on the right. Which of the following will most rapidly decrease this patient's intracranial pressure?

- ☐ A) Trendelenburg (head down) positioning
- ☐ B) Intravenous dexamethasone therapy
- ☐ C) Intravenous mannitol therapy
- ☐ D) Oral glycerol therapy through nasogastric tube
- ☐ E) Intubation and hyperventilation

17. Over the past 6 months, a 60-year-old man has been confused about following directions and has had paresthesias of the hands and feet and an unsteady gait that is worse in the dark. He has atrophic gastritis. Examination shows hyporeflexia of the lower extremities. Babinski sign is present bilaterally. Sensation to vibration and position is markedly decreased over the hands and feet. On mental status testing, he cannot recall three objects after 3 minutes. Which of the following is the most appropriate next step in diagnosis?

- | | |
|--|---|
| ☐ A) CT scan of the head | ☐ F) Measurement of serum electrolyte concentrations |
| ☐ B) HIV antibody test | ☐ G) Measurement of serum thyroid-stimulating hormone concentration |
| ☐ C) Measurement of blood alcohol concentration | ☐ H) Measurement of serum vitamin B ₁₂ (cobalamin) concentration |
| ☐ D) Measurement of erythrocyte sedimentation rate | ☐ I) Serologic test for syphilis |
| ☐ E) Measurement of serum copper concentration | ☐ J) No further testing |

18

A previously healthy 22-year-old woman comes to the emergency department 6 hours after the onset of fever, chills, severe headache, and nausea. Her temperature is 38.5°C (101.3°F), pulse is 100/min, and blood pressure is 100/70 mm Hg. Examination shows nuchal rigidity. There is a petechial rash on the trunk. A lumbar puncture is performed, and antibiotic therapy is begun. After 12 hours, culture of cerebrospinal fluid grows *Neisseria meningitidis*. Which of the following is the most appropriate next step in management for staff who have been in contact with the patient?

- A) Measurement of serum immunoglobulin concentrations
- B) Nasal cultures for *N. meningitidis*
- C) Antibiotic prophylaxis
- D) Antibiotic therapy if symptoms develop
- E) Intramuscular immune globulin therapy

19

A 22-year-old woman is admitted to the hospital with 18 other victims after exposure from an aerial disbursement of an unknown substance. Twelve hours later, she has diplopia, dysphagia, dysphonia, and slight rotatory nystagmus; her gait is unsteady. Her symptoms progress to include flaccid motor paralysis starting at the neck and shoulders downward. The pupils are 6 mm and react sluggishly to light. Deep tendon reflexes are decreased and Babinski sign is absent. Sensation to pinprick is normal throughout. A lumbar puncture is performed. Cerebrospinal fluid analysis shows no abnormalities. Which of the following is the most likely cause of this patient's condition?

- A) Botulism
- B) Exposure to aerosolized snake venom
- C) Exposure to soman nerve gas
- D) Guillain-Barré syndrome
- E) Myasthenia gravis

20

An 82-year-old woman with mild dementia, Alzheimer type, is admitted to the hospital for treatment of pyelonephritis. She has been living at home under the care of her husband. She has been receiving intravenous antibiotics for 1 day. Her pulse is 90/min, respirations are 16/min, and blood pressure is 152/70 mm Hg. Physical examination shows right costovertebral angle tenderness. On mental status examination, she is alert and oriented to person and place but not to time. She is calm. Her husband is concerned because she becomes more confused and agitated when she is away from home, especially at night. Which of the following is most appropriate to prevent agitation in this patient during hospitalization?

- A) Bright illumination of the room at all times
- B) Decrease ambient noise and number of interruptions at night
- C) Hourly orientation by nurses to her situation
- D) Low-dose haloperidol therapy
- E) Use of restraints as needed

21

A previously healthy 50-year-old man comes to the physician because of sinus pressure and purulent nasal discharge for 2 weeks and double vision and partial loss of vision on the right for 18 hours. Three weeks ago, he had an upper respiratory tract infection that resolved spontaneously. Examination shows proptosis, ophthalmoplegia, and partial loss of vision on the right. Which of the following is the most likely diagnosis?

- A) Cavernous sinus thrombosis
- B) Cholesteatoma
- C) Multiple sclerosis
- D) Subdural empyema
- E) Tuberculous meningitis

22

A 37-year-old woman comes to the physician because of a 2-month history of pain, tingling, and numbness in her left wrist radiating to the first three fingers. The symptoms increase over the course of the day. Her job involves typing for several hours daily. Use of a wrist splint at work and at night has provided only minimal relief of her symptoms. Consistent use of high doses of ibuprofen for the past 4 months has provided only partial relief. She is 168 cm (5 ft 6 in) tall and weighs 68 kg (150 lb); BMI is 24 kg/m². Vital signs are within normal limits. Examination shows no enlargement of the thyroid gland. Tapping the volar surface of the left wrist produces an electric shock-like sensation in the hand. Muscle strength and sensation are intact. Which of the following is the most appropriate next step in management?

- A) Glucosamine and chondroitin supplementation
- B) Two-week course of celecoxib twice daily
- C) Two-week course of prednisone
- D) Vitamin B₁₂ (cyanocobalamin) supplementation
- E) Injection of triamcinolone into the carpal tunnel

23

A 42-year-old woman is brought to the emergency department because of a 4-day history of left eye pain and decreasing vision. There is no history of trauma or similar symptoms. Four years ago, she had an episode of severe vertigo, left arm clumsiness, and difficulty walking that spontaneously resolved after 3 weeks. Visual acuity is 20/200 on the left and 20/20 on the right. Examination shows a central scotoma and afferent pupillary defect in the left eye; funduscopic examination shows pallor of the left optic disc. Finger-nose testing shows dysmetria on the left. There is a right pronator drift and right ankle clonus. Babinski sign is present bilaterally. Which of the following is the most appropriate next step in diagnosis?

- A) Measurement of serum Lyme (*Borrelia burgdorferi*) antibody concentration
- B) Carotid duplex ultrasonography
- C) MRI of the brain
- D) Brain stem visual evoked potentials
- E) Lumbar puncture

24. A 12-year-old boy is brought to the physician because of increasing pain in his back and both of his heels since he began playing for his middle school basketball team 3 weeks ago. He also has had difficulty running and jumping because of the pain. He has not had fever. Treatment with ibuprofen has provided minimal relief. His father and paternal uncle have a history of low back pain. The patient is alert and active. He is at the 3rd percentile for height and weight. Examination of the spine shows no deformities or tenderness, but he has limited forward flexion and cannot touch his toes. Dorsiflexion of the ankles is limited to 70 degrees. There is tenderness over the heels bilaterally at the insertion site for the Achilles tendon. Range of motion is otherwise full, and there is no joint pain or swelling. Muscle strength is normal and symmetric in all extremities. Deep tendon reflexes are 2+ in the upper and lower extremities. His gait is normal. The remainder of the examination shows no abnormalities. His erythrocyte sedimentation rate is 55 mm/h. Which of the following is the most likely cause of this patient's heel pain?

- ☐ A) Ankylosing spondylitis
- ☐ B) Growing pains
- ☐ C) Inappropriate arch support
- ☐ D) Pauciarticular juvenile arthritis
- ☐ E) Tethered spinal cord

25. A previously healthy 67-year-old man comes to the physician because of a 2-month history of back pain that awakens him at night. Physical examination, including rectal examination, shows no abnormalities. X-rays of the spine show multiple osteoblastic lesions; a biopsy specimen of the lesions shows well-differentiated adenocarcinoma. Which of the following is the most appropriate next step to establish the origin of cancer in this patient?

- ☐ A) Measurement of serum α -fetoprotein concentration
- ☐ B) Measurement of serum prostate-specific antigen concentration
- ☐ C) Urine immunoelectrophoresis
- ☐ D) X-ray of the chest
- ☐ E) Colonoscopy

26. A 47-year-old woman is brought to the emergency department 4 hours after the sudden onset of difficulty with speech and paralysis on the right. She has had fever, chills, and anorexia for 2 weeks. She has a history of rheumatic heart disease. Her temperature is 39.5°C (103.1°F). A loud systolic murmur is heard. Examination shows weakness and numbness of the right side of the face and the right upper extremity, with relative sparing of the right lower extremity. Her speech is markedly nonfluent, but comprehension is relatively intact. A CT scan of the head without contrast shows normal findings. Echocardiography shows a vegetation on the mitral valve. Blood has been drawn for culture. Which of the following is the most appropriate next step in management?

- ☐ A) Antibiotic therapy
- ☐ B) β -Adrenergic blocking agent therapy
- ☐ C) Burr hole
- ☐ D) Corticosteroid therapy
- ☐ E) Duplex scan of the carotid arteries
- ☐ F) Heparin therapy
- ☐ G) Lumbar puncture
- ☐ H) MRI of the brain

27. A 32-year-old woman with AIDS is brought to the physician by her husband because of bizarre behavior over the past week. He says that she has not gone to work for the past 3 days, but she has been going to the movies instead and sitting through the same movie two or three times in a row. Physical examination shows lymphadenopathy. Neurologic examination shows hyperreflexia of the right upper and lower extremities. Language function is normal, but she is tangential and perseverative in conversation. A CT scan of the head shows a region of hypodensity in the left putamen with ring enhancement after contrast administration. There are two similar smaller lesions in the right frontal lobe and the left midbrain. Which of the following is the most likely causal organism?

- ☐ A) *Cryptococcus neoformans*
- ☐ B) Cytomegalovirus
- ☐ C) Herpes simplex virus
- ☐ D) *Listeria monocytogenes*
- ☐ E) *Mycobacterium avium-intracellulare*
- ☐ F) *Toxoplasma gondii*
- ☐ G) *Treponema pallidum*

28. A 37-year-old man is brought to the emergency department by a coworker 30 minutes after having a generalized tonic-clonic seizure. They were picking fruit in a field that was recently sprayed with an unknown substance when the patient had the onset of abdominal cramps and blurred vision and then had the seizure. On arrival, he is drowsy but arousable. He is salivating and clutching his abdomen. His temperature is 37°C (98.6°F), pulse is 40/min, respirations are 22/min, and blood pressure is 90/60 mm Hg. Examination shows profuse diaphoresis and diffuse muscle twitching. The pupils are 1 mm and react sluggishly to light. There is no ptosis. Abdominal examination shows no organomegaly. Which of the following is the most likely mechanism of these findings?

- ☐ A) Decreased adrenergic activity
- ☐ B) Decreased cholinergic activity
- ☐ C) Decreased dopaminergic activity
- ☐ D) Decreased serotonergic activity
- ☐ E) Increased adrenergic activity
- ☐ F) Increased cholinergic activity
- ☐ G) Increased dopaminergic activity
- ☐ H) Increased serotonergic activity

29. A 42-year-old man comes to the physician because of a 2-week history of pain in the right side of the neck that radiates down the lateral aspect of the right arm to the third digit. He first noted the pain after lifting his 5-year-old son. He has also had mild weakness of the right arm and tingling in the right third digit. He works as a computer programmer, and spends most of his day at a keyboard. Examination shows mild weakness of right elbow and wrist extension, and a decreased right triceps reflex. Sensory examination shows no abnormalities. Which of the following is the most likely location of the lesion causing this patient's symptom?

- ☐ A) Basal ganglia
- ☐ B) Brachial plexus
- ☐ C) Cervical nerve root
- ☐ D) Cervical spinal cord
- ☐ E) Median nerve
- ☐ F) Radial nerve
- ☐ G) Thoracic nerve root
- ☐ H) Ulnar nerve

30. A 19-year-old man is brought to the emergency department by his father 30 minutes after he found his son lying in bed unresponsive to his words or to shaking. His father says his son felt hot to the touch. Two days ago, the patient began risperidone therapy after an episode of agitation and visual hallucinations. The patient has become increasingly withdrawn since he took his first dose of risperidone. He has no other history of medical or psychiatric illness. His temperature is 40°C (104°F), pulse is 110/min, respirations are 15/min, and blood pressure is 150/90 mm Hg. His eyes are open; the pupils are 4 mm and reactive. There is extreme rigidity of all extremities. He responds sluggishly to painful stimuli. Laboratory studies show a leukocyte count of 11,000/mm³ and serum creatine kinase activity of 1500 U/L. Urinalysis is positive for myoglobin. Dysregulation of which of the following neurotransmitters is most likely responsible for this patient's symptoms?
- ☐ A) γ -Aminobutyric acid
 - ☐ B) Dopamine
 - ☐ C) Glutamate
 - ☐ D) Norepinephrine
 - ☐ E) Serotonin
31. A 22-year-old man with Duchenne muscular dystrophy is brought to the physician by his parents because of increased difficulty clearing secretions over the past 3 weeks. He is confined to bed or a chair and needs assistance with all activities of daily living. He lives with his parents who are upset that he does not want to have friends visit. He says that he is often irritated by their attempts to keep his spirits up and to make him more comfortable. Both the patient and family have avoided discussing the patient's wishes regarding end-of-life care. Vital signs today are within normal limits. Examination shows generalized muscle weakness and wasting; he needs support to remain seated in his wheelchair. Upper airway sounds are heard without the stethoscope. Which of the following is the most appropriate next step in management?
- ☐ A) Clarify the patient's preferences for treatment as his disease progresses
 - ☐ B) Obtain permission for tracheostomy
 - ☐ C) Refer the patient for a psychiatric evaluation
 - ☐ D) Begin fluoxetine therapy
 - ☐ E) Begin nasopharyngeal suctioning with oxygen via nasal cannula as needed

32. A 27-year-old woman comes to the physician requesting a prescription for citalopram because of a long-standing depressed mood and low self-esteem. She reports that she has felt depressed and stressed since returning to work after the birth of her second child 3 years ago. She does not feel that she gets enough help from her husband. She also has had decreased energy and has lost interest in things she used to enjoy. She has not had changes in sleep or appetite. She has no history of serious illness and takes no medications. Physical examination shows no abnormalities. On mental status examination, she is engaging with a full range of affect. There is no evidence of psychosis or suicidal ideation. Which of the following is the most likely diagnosis?
- ☐ A) Adjustment disorder
 - ☐ B) Dysthymic disorder
 - ☐ C) Major depressive disorder
 - ☐ D) Occupational problem
 - ☐ E) Partner relational problem
33. A 77-year-old man is brought to the physician by his wife because of a 6-month history of memory problems, frequent falls, and urinary incontinence. His wife says that he "shuffles" when walking and falls forward or backward when getting out of a chair. He has gotten lost several times when driving to the local grocery store and could not complete the family taxes this year. He has hypertension well controlled with lisinopril. His temperature is 37°C (98.6°F), pulse is 90/min and regular, and blood pressure is 135/80 mm Hg. Muscle strength and tone are normal. Deep tendon reflexes are 2+ throughout. Babinski sign is absent. His gait is narrow based with small steps; he takes eight steps to turn 180 degrees. Arm swing is normal. His Mini-Mental State Examination score is 22/30. Which of the following is the most likely diagnosis?
- ☐ A) Brain tumor
 - ☐ B) Normal-pressure hydrocephalus
 - ☐ C) Parkinson disease
 - ☐ D) Subdural hematoma
 - ☐ E) Vitamin B₁₂ (cobalamin) deficiency
34. A 57-year-old woman comes to the physician because of a 6-week history of progressive hearing loss in her right ear. Five years ago, she underwent a successful surgical removal of a left acoustic neuroma (vestibular schwannoma). Her temperature is 37°C (98.6°F). Physical examination, including examination of the skin and the tympanic membranes, shows no abnormalities. Neurologic examination shows decreased hearing in the right ear; hearing on the left is normal. Air conduction is greater than bone conduction bilaterally. A tuning fork held over the forehead lateralizes to the left ear. Which of the following is the most likely diagnosis?
- ☐ A) Cholesteatoma
 - ☐ B) Neurofibromatosis
 - ☐ C) Otitis media
 - ☐ D) Otosclerosis
 - ☐ E) Presbycusis

35. A 52-year-old man had the acute onset of right eye pain and global headache 8 hours ago and markedly blurred vision in the right eye 2 hours ago. Examination of the right eye shows diffuse redness of the conjunctiva with circumcorneal prominence; the pupil is midrange and nonreactive to light. There is no exudate. Which of the following is the most appropriate next step in management?

- ☐ A) CT scan of the head
- ☐ B) 5% Carbon dioxide therapy
- ☐ C) Opiate analgesia therapy
- ☐ D) Topical corticosteroid therapy
- ☐ E) Tonometry

36. A 3-month-old boy is brought to the emergency department 20 minutes after a generalized tonic-clonic seizure that lasted 5 minutes. He has had low-grade fever, intermittent vomiting, and diarrhea for the past 2 days; he has been treated with acetaminophen for the fever. He has refused formula but has taken glucose and water over the past 10 hours; he has not urinated during this period. An older sibling had a similar illness 1 week ago. He is quiet but easily arousable. His temperature is 37°C (98.6°F), pulse is 160/min, respirations are 36/min, and blood pressure is 60/40 mm Hg. Examination shows a sunken anterior fontanelle and eyes and dry mucous membranes. The remainder of the examination shows no abnormalities. Which of the following is most likely to confirm the cause of the seizure?

- ☐ A) Culture of cerebrospinal fluid
- ☐ B) Measurement of serum calcium concentration
- ☐ C) Measurement of serum creatinine concentration
- ☐ D) Measurement of serum sodium concentration
- ☐ E) Metabolic screening
- ☐ F) Toxicology screening

37. A 62-year-old woman with severe rheumatoid arthritis is admitted to the hospital 24 hours before she is scheduled to undergo elective right total knee replacement. Her arthritic symptoms are currently well controlled with methotrexate therapy. She has not taken corticosteroid therapy for more than 2 weeks during the past year. Vital signs are within normal limits. Cardiopulmonary examination shows no abnormalities. Examination of the upper extremities shows marked deformities of the hands with swelling deformity of the metacarpophalangeal joints and the proximal interphalangeal joints. The hands have ulnar deviation. There are several rheumatoid nodules on the upper extremities. Examination of the knees shows bilateral degenerative changes that are worse on the right; there is synovial thickening but no erythema or effusion. Rectal examination shows poor sphincter tone but no other abnormalities. Which of the following studies is most appropriate before surgery?

- ☐ A) Serum rheumatoid factor assay
- ☐ B) Overnight dexamethasone suppression test
- ☐ C) Measurement of serum cortisol concentrations immediately before and 30 minutes and 60 minutes after high-dose cosyntropin injection
- ☐ D) Flexion and extension x-rays of the cervical spine
- ☐ E) Pulmonary function testing

38. A 72-year-old man comes to the physician because he has had several 4-minute episodes of twitching of the left arm since discharge from the hospital 3 weeks ago following a right cerebral hemisphere infarction. Examination shows no abnormalities except for mild left hemiparesis. Which of the following is the most likely cause of this patient's symptoms?

- ☐ A) Absence seizures
- ☐ B) Atonic seizures
- ☐ C) Complex partial seizures
- ☐ D) Myoclonic seizures
- ☐ E) Psychogenic seizures
- ☐ F) Simple partial seizures

39. A 77-year-old woman comes to the physician because of several episodes of syncope during the past 6 months. She has not noticed any precipitating events that cause the episodes. The episodes usually resolve within 3 minutes without sequelae. She is otherwise asymptomatic. She has hypertension and hypercholesterolemia. Current medications include metoprolol and atorvastatin. Her temperature is 37°C (98.6°F), pulse is 60/min, respirations are 12/min, and blood pressure is 160/95 mm Hg in the right arm and 90/60 mm Hg in the left arm. Physical examination, including neurologic examination, shows no abnormalities. Which of the following is the most likely underlying mechanism of syncope in this patient?

- ☐ A) Brain stem embolization due to innominate artery stenosis
- ☐ B) Decrease in brain stem perfusion due to reversal of flow in one vertebral artery
- ☐ C) Decrease in cardiac output due to spontaneous cardiac arrhythmias
- ☐ D) Intermittent vasovagal reactions
- ☐ E) Seizure disorder

During a phlebotomy, a 35-year-old woman becomes pale, diaphoretic, and light-headed and then loses consciousness. There is stiffening of all extremities for a few seconds. Within 45 seconds of being moved to a supine position, she awakens and is weak and vomits. Her pulse is 80/min, respirations are 16/min, and blood pressure is 126/82 mm Hg. The remainder of the physical and neurologic examinations is normal. Which of the following is the most likely explanation for this episode?

- A) Complex partial seizure
- B) Generalized tonic-clonic seizure
- C) Hypovolemia
- D) Third-degree atrioventricular block
- E) Vasovagal attack

41. A 77-year-old woman is brought to the emergency department 5 hours after the onset of dizziness, blurred vision, numbness of the right side of her face, and slurred speech. She has had three similar 30- to 40-minute episodes during the past 3 weeks. She feels well between episodes. She frequently has tinnitus in her left ear caused by chronic mastoiditis and neck pain caused by osteoarthritis. She undergoes cervical manipulation by a chiropractor four times monthly. She underwent coronary artery bypass grafting 4 years ago. Medications include clopidogrel and isosorbide. Her pulse is 85/min, respirations are 20/min, and blood pressure is 140/104 mm Hg. Examination, including neurologic examination, shows no abnormalities except for a well-healed midline sternal scar. A CT scan of the head and an MRI of the brain show no abnormalities. Magnetic resonance angiography shows dissection of the right vertebral artery. Which of the following is the most appropriate immediate pharmacotherapy?

- ☐ A) Intravenous heparin
- ☐ B) Intravenous tissue plasminogen activator
- ☐ C) Intravenous urokinase
- ☐ D) Oral dipyridamole
- ☐ E) Oral warfarin

42. A previously healthy 11-year-old girl is brought to the physician by her mother because of a 3-month history of severe daily headaches, blurred vision, and urinary frequency. Menarche has not yet begun. She is at the 20th percentile for height and 75th percentile for weight. Vital signs are within normal limits. Visual acuity is 20/20 bilaterally. Examination shows bitemporal visual field defects, papilledema, and bilateral weakness of eye abduction. A CT scan of the head shows a large suprasellar mass containing numerous specks of calcium. Which of the following is the most likely diagnosis?

- ☐ A) Craniopharyngioma
- ☐ B) Idiopathic intracranial hypertension
- ☐ C) Medulloblastoma
- ☐ D) Migraine
- ☐ E) Parasagittal meningioma
- ☐ F) Pituitary adenoma
- ☐ G) Subdural hematoma
- ☐ H) Temporal lobe glioma

43. A previously healthy 62-year-old man is brought to the physician by his wife because of confusion for 3 hours. During breakfast, he stopped eating and asked his wife the day's date and plans for the day. He repeated his questions six times despite receiving the correct information from his wife each time. He now is oriented and clear in thinking. He does not recall any aspect of the morning since breakfast. Physical and mental status examinations show no abnormalities. Which of the following is the most likely diagnosis?

- ☐ A) Communicating hydrocephalus
- ☐ B) Concussion
- ☐ C) Dementia, Alzheimer type
- ☐ D) Herpes simplex encephalitis
- ☐ E) Hypoxic-ischemic encephalopathy
- ☐ F) Hysterical amnesia
- ☐ G) Transient global amnesia
- ☐ H) Wernicke-Korsakoff syndrome

44. A 62-year-old man has the sudden onset of numbness and weakness of the right arm and face accompanied by difficulty with speech that resolves completely within 30 minutes. He has a 15-year history of hypertension. His pulse is 76/min and regular, and blood pressure is 150/85 mm Hg. Bilateral carotid bruits are heard. Cardiac and neurologic examinations show normal findings. An ECG and echocardiography show normal findings. Bilateral carotid angiography shows 40% stenoses in both internal carotid arteries. Which of the following is the most effective measure to prevent a cerebral infarction in this patient?

- ☐ A) Administration of aspirin
- ☐ B) Administration of low-molecular-weight heparin
- ☐ C) Administration of warfarin
- ☐ D) Left carotid endarterectomy
- ☐ E) Bilateral carotid endarterectomy

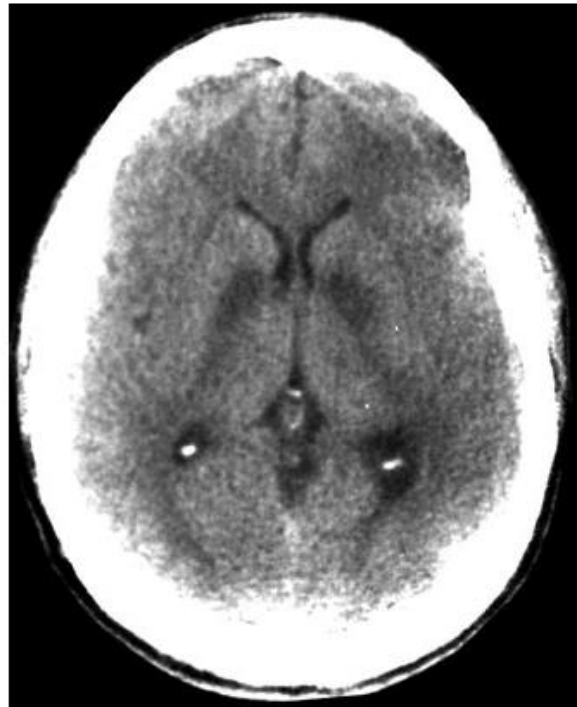
45. A 67-year-old man comes to the physician because of a 4-month history of falls, which have increased in frequency during the past month. He has a long-standing history of abdominal bloating after most meals. He has type 2 diabetes mellitus, coronary artery disease, and osteoporosis. Medications are alendronate, metformin, metoclopramide, and transdermal nitroglycerin. His blood pressure is 140/88 mm Hg while supine, 125/90 mm Hg while sitting, and 120/92 mm Hg while standing. Physical examination shows diminished facial expression, infrequent blinking, and a hypophonic voice. Spontaneous movement is decreased in all extremities. Cogwheel rigidity is noted at both wrists. On gait testing, he walks in short steps. Mental status examination shows no abnormalities. Which of the following is the most appropriate next step in pharmacotherapy?

- ☐ A) Discontinue alendronate
- ☐ B) Discontinue metoclopramide
- ☐ C) Add carbidopa-levodopa to the current medication regimen
- ☐ D) Add ropinirole to the current medication regimen
- ☐ E) Add trihexyphenidyl to the current medication regimen

46. A 15-year-old boy is brought to the physician by his mother because of poor academic performance since he began 10th grade 2 months ago. His teachers are concerned he does not get sufficient sleep because he frequently falls asleep in class. He also has had episodes of acute weakness of his neck and knee muscles that typically occur when he is socializing with friends. He is 163 cm (5 ft 4 in) tall and weighs 75 kg (165 lb); BMI is 28 kg/m². Physical examination shows no other abnormalities. Which of the following is the most likely diagnosis?

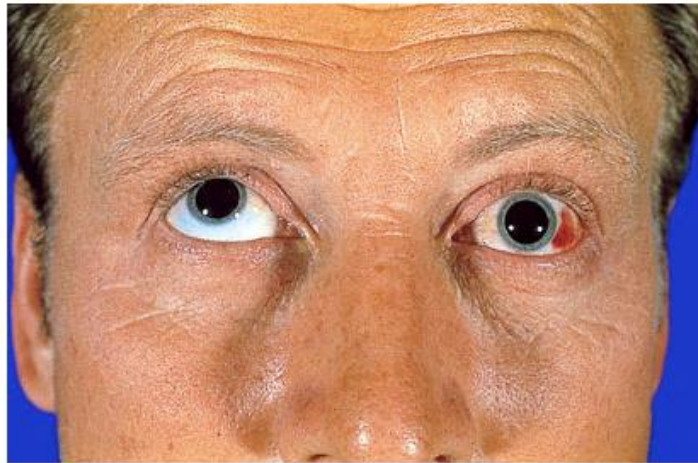
- ☐ A) Delayed sleep phase syndrome
- ☐ B) Kleine-Levin syndrome
- ☐ C) Narcolepsy
- ☐ D) Obstructive sleep apnea
- ☐ E) Seizure disorder

47. A 27-year-old woman is brought to the emergency department in winter by a neighbor 15 minutes after being found unconscious at home. Her neighbor reports that she had not seen the patient for 2 days. The patient has type 1 diabetes mellitus. She lives alone and has been refurbishing her house, which she recently purchased. On arrival, she is comatose. Vital signs are within normal limits. The pupils are 4 mm and reactive to light. On examination, she does not respond to stimuli. Babinski sign is present bilaterally. Results of laboratory studies are pending. A CT scan of the head without contrast is shown. Which of the following is the most likely diagnosis?



- ☐ A) Aspirin overdose
- ☐ B) Carbon monoxide poisoning
- ☐ C) Hypercalcemia
- ☐ D) Ketoacidosis
- ☐ E) Lead poisoning

48. A 42-year-old man comes to the physician because of double vision. Two days ago, he was hit in the left eye with a softball. There was no loss of consciousness, but he was unable to continue playing. He took aspirin and rested yesterday. Examination is normal except when he is asked to look up. A photograph is shown. Which of the following is the most likely cause of the diplopia?



- ☐ A) Entrapment of inferior oblique
- ☐ B) Orbital hemorrhage
- ☐ C) Paresis of the sixth cranial nerve
- ☐ D) Paresis of the third cranial nerve
- ☐ E) Subdural hematoma

49. A 47-year-old woman comes to the physician because she has been binge drinking for the past 3 days; her last drink was 4 hours ago. She had been abstinent for the past month. She says that she drank socially during her 20s and early 30s, but she started to drink regularly and more heavily beginning in her late 30s. She has no history of delirium tremens or withdrawal seizures. She has panic disorder treated with paroxetine. Her father and brother have received treatment for substance abuse. She says that she has a severe headache. She appears diaphoretic. Her temperature is 36.9°C (98.4°F), pulse is 110/min, respirations are 18/min, and blood pressure is 165/105 mm Hg. Physical examination shows a moderate tremor of the upper extremities. On mental status examination, she is moderately anxious, fidgety, and restless. Her thought process is rational, and she is oriented to person, place, and time. Serum studies show an AST activity of 35 U/L and an ALT activity of 30 U/L. Her blood alcohol concentration is 100 mg/dL. In addition to admission to the hospital, which of the following is the most appropriate pharmacotherapy?
- ☐ A) Atenolol
 - ☐ B) Carbamazepine
 - ☐ C) Clonidine
 - ☐ D) Diazepam
 - ☐ E) Risperidone
50. A previously healthy 37-year-old man comes to the emergency department because of a 4-hour history of paralysis of the left side of his face. When he awoke this morning, he noticed a facial droop that has progressed to include his left eyebrow. He takes no medications and does not smoke. He exercises regularly by biking. He is 180 cm (5 ft 11 in) tall and weighs 84 kg (185 lb); BMI is 26 kg/m². His temperature is 37°C (98.6°F), pulse is 56/min, and blood pressure is 110/72 mm Hg. Examination shows loss of the left nasolabial fold. He has difficulty closing the left eyelid. The pupils are equal and reactive to light. Hearing is intact. The tympanic membranes and pharynx appear normal, and the remainder of the examination shows no abnormalities. Which of the following is the most appropriate next step in diagnosis?
- ☐ A) MRI of the brain
 - ☐ B) Tensilon test
 - ☐ C) Lumbar puncture
 - ☐ D) Nerve conduction studies
 - ☐ E) No further testing is indicated